

Predicting treated-water quality

Iteration 1 APPLIES RUN 1'S NEXT INVESTIGATIONS

GUARDRAILS · CHECKED BY THE RUN

- ✓ Leakage: no feature measured with the outcome
- ✓ Split: every test day later than every training day
- ✓ Baseline computed first (8.39); MAE 4.04 reported against it
- ✓ Imputation: training-set medians only, columns listed below
- ✓ Target MAE ≤ 4.5 : **reached at 4.04** — ticked by the run

NEXT INVESTIGATIONS · THE SUGGESTIONS

- Add **load = BOD × solids** — a chemist's feature, not a bigger model.
- Reframe as a limit alarm (classification) and price false positives against false negatives.
- Re-evaluate monthly for drift; retrain when MAE approaches the baseline gap.

The first line is what **run 3** executes.

Iterated from run 1: time-aware split adopted · DQO-S removed. The score fell from 0.868 to 0.706 — the drop *is* the correction.

1 Question

Predict tomorrow's **output BOD (DBO-S, mg/L)** for the treatment plant, from today's inflow measurements and settings. Success metric: **MAE**. The baseline to beat: predict the historical mean.

2 Data

527 daily records, 12 variables, 92 missing cells (imputed with training-set medians). Schema follows the UCI Water Treatment Plant dataset (Poch 1993, CC BY 4.0); values regenerated offline for this session.

3 Features

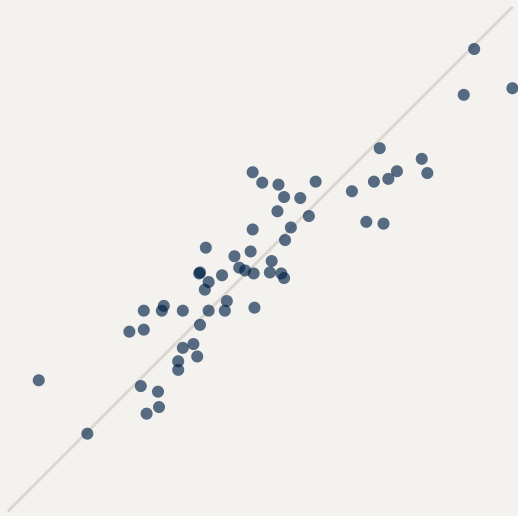
Q-E, PH-E, DBO-E, DQO-E, SS-E, SED-E, COND-E

4 Model & split

Linear regression. Split: **time-aware (last 20% of days)**.

5 Evaluation

MAE on test	4.04 mg/L
R ² on test	0.706
Baseline MAE (predict the mean)	8.39 mg/L



predicted vs actual, test set (perfect = diagonal)

✓ **Human validation — to be ticked by the signatory**

- ☐ The question is stated with a target and a unit
- ☐ Every feature exists at prediction time (no leakage)
- ☐ The split respects time: test days are strictly later
- ☐ The metric is reported on the test set only, next to the baseline
- ☐ Worst errors inspected; limits and uncertainty stated
- ☐ I approve this report for use within its stated limits

All runs precomputed for the session (no live computation on stage). Skill: skills/ml-analysis – swap in the real water-treatment.data with one line. Sources: UCI id 106; Sculley et al., NeurIPS 2015.

GUARDRAILS – THE NUMBERED LIST FROM PROMPT.TXT, VERIFIED BY THE RUN

G1	PASS	Target variable and unit stated — DBO-S, mg/L
G2	PASS	Imputation applied, training-set medians; imputed columns listed above
G3	PASS	No feature measured with or after the outcome — DQO-S excluded
G4	PASS	Every test day later than every training day; no shuffle
G5	PASS	Baseline computed first (8.39) and printed beside every score
G6	PASS	Limits and uncertainty stated
G7	PASS	Target reached — $MAE\ 4.04 \leq 4.5$, and G1–G6 all pass
H1	HUMAN	Human sign-off — left empty by the run

Scientist’s panel — read before signing

ITERATION 1 · APPLIES RUN 1’S NEXT INVESTIGATIONS

Iterated from run 1: time-aware split · DQO-S removed.

GUARDRAILS CHECKLIST

- ✓ No feature measured with the outcome (DQO-S excluded)
- ✓ Time-aware split: every test day later than every training day
- ✓ Baseline computed first; every score reported next to it (8.39)
- ✓ Imputation: training-set medians only, columns listed above
- ✓ Target MAE ≤ 4.5 — reached at 4.04 · ticked by the run

LIMITS & UNCERTAINTY

Valid for this plant and this period only. MAE 4.04 mg/L is an average: worst days reach three times that. Not tested across seasons or after process changes (drift).

NEXT INVESTIGATIONS

- Add the load feature (BOD × solids) — domain knowledge, tested in run 3.
- Reframe as a limit alarm (classification) and price false positives vs false negatives.
- Re-evaluate monthly against drift; retrain when MAE degrades past the baseline gap.

Sign-off (human only): approved by _____ date _____